



Homework 1

EX2.

by following exercise, we have $K2 \iff P2$ and $K4 \iff P4$, therefore, the total axioms we can use are:

$$K2: KE \subseteq E$$

$$P2: E \subseteq PE$$

$$K4: PE \subseteq KPE$$

$$P4: PKE \subseteq KE$$

besides, by $KPE = K(PE) \subseteq PE$ and $PKE = P(KE) \supseteq KE$, we can prove the sign of inequality in K4/P4 can be replaced by sign of equality, and we will directly use the equality relation when we applying K4 or P4 at above calculation.

$$KE \xrightarrow{P2} PKE \xrightarrow{K4} KPKE \xrightarrow{P4} KKE = K^2E$$

$$\implies KE \subseteq K^2E$$

EX3.

- K0/P0

$$K\Omega = \Omega \iff K \sim \emptyset = \sim \emptyset \iff \sim K \sim \emptyset = P\emptyset = \sim \sim \emptyset = \emptyset$$

- K1/P1

$$K(E \cap F) = KE \cap KF \iff K(\sim E) \cap (\sim F) = (K \sim E) \cap (K \sim F)$$

$$\iff \sim K[(\sim E) \cap (\sim F)] = \sim K[\sim (E \cup F)] = \sim [(K \sim E) \cap (K \sim F)] = \sim$$

$$K \sim E \cup \sim K \sim F$$

$$\iff P(E \cup F) = PE \cup PF$$

- K2/P2

$$Ke \subseteq E \iff K \sim E \subseteq \sim E \iff \sim K \sim E = PE \supseteq E = \sim \sim E$$

- K3/P3

$$KE \subseteq E \iff K \sim E \subseteq \sim E \iff \sim K \sim E \supseteq \sim \sim E \iff PE \supseteq E$$

- K4/P4

$$PE \subseteq KPE \iff P \sim E \subseteq KP \sim E \iff \sim KE \subseteq K \sim KE \iff KE \supseteq \sim K \sim KE = PKE$$

EX4.

- a. $K(KE) = KE$ from K3 and the textbook supplement
- b. $K \sim KE = K(P \sim E) \stackrel{P3}{=} P \sim E = \sim KE$
- c. $KPE = PE$ from P4 and textbook supplement
- d. $K \sim PE = KK \sim E \stackrel{K3}{=} K \sim E = \sim PE$

EX5.

people	state of Dirty face
Alice	2,5,6,8
Beatrice	3,5,7,8
Carol	4,6,7,8

initial

people	partition
Alice	$\{1, 2\}, \{3, 5\}, \{4, 6\}, \{7, 8\}$
Beatrices	$\{1, 3\}, \{2, 5\}, \{4, 7\}, \{6, 8\}$
Carol	$\{1, 4\}, \{2, 6\}, \{3, 7\}, \{5, 8\}$

after the broadcast:

people	partition
Alice	$\{1\}, \{2\}, \{3, 5\}, \{4, 6\}, \{7, 8\}$
Beatrices	$\{1\}, \{3\}, \{2, 5\}, \{4, 7\}, \{6, 8\}$

people	partition
Carol	$\{1\}, \{4\}, \{2, 6\}, \{3, 7\}, \{5, 8\}$

after the first second

Alice knowing the partition of B and C

- B blurred if and only if the true state is 3
- C blurred if and only if the true state is 4

Therefore, A can determine if the state is 4 and 3 by observing the face of B and C with the same method,

- B can determine if the state is 2 or 4
- C can determine if the state is 2 or 3

Therefore, the partition after 1 second will become

people	partition
Alice	$\{1\}, \{2\}, \{3\}, \{5\}, \{4\}, \{6\}, \{7, 8\}$
Beatrices	$\{1\}, \{3\}, \{2\}, \{5\}, \{4\}, \{7\}, \{6, 8\}$
Carol	$\{1\}, \{4\}, \{2\}, \{6\}, \{3\}, \{7\}, \{5, 8\}$

after the second second

After the first second, A would able to determine all states except 7 and 8.

By the information of B and C , A will find that

- if the true state is 3, 5, 7, then B will blurred
- if the true state is 4, 6, 7, then C will blurred

Therefore, A will able to if the state is 7 or 8 by observing the face of B or C

Therefore, A will able to distinguish all states after the 2nd second.

The same thinking process will applied to B and C

Therefore, the partition will become as follow:

people	partition
Alice	$\{1\}, \{2\}, \{3\}, \{5\}, \{4\}, \{6\}, \{7\}, \{8\}$
Beatrices	$\{1\}, \{3\}, \{2\}, \{5\}, \{4\}, \{7\}, \{6\}, \{8\}$
Carol	$\{1\}, \{4\}, \{2\}, \{6\}, \{3\}, \{7\}, \{5\}, \{8\}$

in other words, after 2 seconds, all people will know if their face is dirty.

EX6.

a.

a Truism is $\emptyset$

b.

A cannot know $\{4, 5\}$ at each event:

from the topic given, we have

- $K_A\{1, 2, 4, 5\} = \{1, 2\}$
- $K_A\{2, 3, 4, 5\} = \{3, 4, 5\}$

$$\implies K_A\{2, 4, 5\} = \emptyset$$

from K2, we can find that if $E_1 \subseteq E_2$, then $KE_1 = K(E_1 \cap E_2) = KE_1 \cap KE_2 \subseteq E_2$

$$\implies K_A\{4, 5\} \subseteq K_A\{2, 4, 5\} = \emptyset$$

In other words, A cannot know $\{4, 5\}$ whatever the true state is.

c.

the possibility set partition of B is

$\{1, 2, 3\}, \{4, 5\}$,

if the true state is located in $\{1, 2\}$

- at this state, they can only find the true state located in $P_A\{1, 2\} \cap P_B\{1, 2\} = \{1, 2\} \cap \{1, 2, 3\} = \{1, 2\}$

if the true state is located is 3

- at this state, $P_A(3) = \{3, 4, 5\}$, and B will give the information that the true state is located in $P_B(3) = \{1, 2, 3\}$, then A will find the true state is located in $P_A(3) \cap P_B(3) = \{3\}$.
in other word, A will determine the true state is 3

if the true state is located in $\{4, 5\}$

- they will know the true state is in $P_A\{4, 5\} \cap P_B\{4, 5\} = \{3, 4, 5\} \cap \{4, 5\} = \{4, 5\}$

Therefore, the Possibility set partition of A will become $\{1, 2\}, \{3\}, \{4, 5\}$